Removing Antipodality from Asymptotically Optimal Partial Sphere Coverings

Literature-implied solution packet

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Status. This is a full answer to one open direction in Hoehner–Kur, arXiv:2501.10607, but it is classified as *literature-implied*, not as a wholly original proof. The decisive Gaussian comparison is Theorem 2.1 of Mulgund, arXiv:2607.14087. The identification of that theorem with the source paper's cone–halfspace reduction appears to be new; no explicit public linkage was found.

1 Source question and exact scope

Let $C_d(x)$ be the geodesic cap in $\mathbb{S}^{d-1}$ centered at $x \in \mathbb{S}^{d-1}$ and having normalized surface measure $1/N$. Define the unrestricted optimum

$$U_N(d) := \sup_{x_1,\dots,x_N \in \mathbb{S}^{d-1}} \sigma_{d-1}\left(\bigcup_{i=1}^{N} C_d(x_i)\right).$$

Hoehner–Kur prove the limit $1-e^{-1}$ when the centers occur in antipodal pairs. On page 2 they explicitly write that they "do not know how to remove the antipodality assumption." The surrounding paragraph is reproduced below for location verification.

Equation (2) shows that, like π, Euler's number e may also be defined in terms of the geometry of the sphere in high dimensions. Theorem 1 also confirms [18, Conjecture 4.2] in the case of antipodal partial coverings by $N = N(d)$ caps satisfying $N(d) \to \infty$ and $\ln N(d) = o(\sqrt{d})$, showing that, under these assumptions, random partial coverings are indeed asymptotically optimal (up to a $1 + o(1)$ factor).

To establish the equality in Theorem 1, we show the inequality in both directions. For the lower bound, we use random antipodal partial coverings. We follow the arguments in [18], with slight modifications made since our partial coverings are antipodal. We refer the reader to [18, Subsection 9.1] for the details. Thus, the main contribution of this paper is the upper bound in Theorem 1.

Nevertheless, the present paper leaves several open directions. In particular, we do not know how to remove the antipodality assumption, or how to extend the result to the regime $N = e^{\Omega(\sqrt{d})}$. Our method also does not yet resolve whether $V_N(d)$ decreases monotonically to $1 - e^{-1}$, a property that would describe how overlaps grow with increasing dimension.

Remark 1. In [14], Erdős, Few and Rogers study the following asymptotic version of the partial covering problem. For each $r > 0$, consider a finite family Σ_r of congruent caps on $r\mathbb{S}^{d-1}$, and suppose that

The present packet settles precisely that first direction in the same regime. It does not settle the paper's exponential-regime, monotonicity, or Laguerre–Dirichlet–Voronoi questions.

2 Result

Theorem 1 (Full non-antipodal extension)**.** *Let $N = N(d)$ be any integer sequence such that $N(d) \to \infty$ and $\log N(d) = o(\sqrt{d})$. Then*

$$\lim_{d\to\infty} U_N(d) = 1 - e^{-1}.$$

Thus arbitrary arrangements of N equal caps are asymptotically no better than random arrangements; no antipodal hypothesis is needed.

The proof imports one later theorem and otherwise uses estimates already proved by the source paper. We state the imported result in exactly the form needed here.

Theorem 2 (Mulgund, Theorem 2.1)**.** *Let R be an $m \times m$ correlation matrix and let $J = \mathbf{1}\mathbf{1}^T$. If $R - J/m \succeq 0$ and $X \sim N(0, R)$, then for every $c \in \mathbb{R}$,*

$$\mathbb{P}\{X_i \le c \text{ for all } i\} \ge \Phi(c)^m.$$

Singular correlation matrices are allowed.

3 Proof intuition

The source paper replaces each spherical cap by its Gaussian cone and then shows that, at a slightly trimmed level u, each cone differs from the one-sided Gaussian halfspace with the same axis by only $o(1/N)$ in measure. This estimate is rotationally uniform and does not use antipodality. Consequently, the unrestricted problem becomes a statement about the maximum of an arbitrary centered Gaussian vector with unit coordinate variances.

Mulgund's theorem does not apply directly to its covariance G. The useful normalization is

$$R = \frac{N-1}{N} G + \frac{1}{N} J.$$

It is always admissible because $R - J/N = ((N-1)/N)G \succeq 0$. Probabilistically, this merely rescales the original Gaussian vector and adds the same small independent normal variable to every coordinate. At an extreme-value level $u \sim \sqrt{2 \log N}$, that common perturbation is negligible. Mulgund's independent lower-orthant bound therefore yields a universal no-hit probability of at least $e^{-1} - o(1)$.

4 Proof

Write $\overline{\Phi}(t) = 1 - \Phi(t)$. We use the following consequence of Theorem 2.

Lemma 1 (Universal rare-exceedance bound)**.** *For each N, let G_N be an arbitrary $N \times N$ correlation matrix and let $\xi^{(N)} \sim N(0, G_N)$. If $u_N \to \infty$ and $N\overline{\Phi}(u_N) \to 1$, then, uniformly over G_N,*

$$\mathbb{P}\left\{\max_{1\le i\le N} \xi_i^{(N)} \le u_N\right\} \ge e^{-1} - o(1).$$

Proof. Set $a_N = (N-1)/N$ and

$$R_N = a_N G_N + \frac{1}{N} J.$$

This is a correlation matrix and $R_N - J/N = a_N G_N \succeq 0$. If W is an independent standard normal random variable, then

$$X_i := \sqrt{a_N}\,\xi_i^{(N)} + \frac{W}{\sqrt{N}}$$

has covariance R_N. Let $L_N = N^{1/4}$ and

$$c_N := \sqrt{a_N}\,u_N - \frac{L_N}{\sqrt{N}} = \sqrt{a_N}\,u_N - N^{-1/4}.$$

Because the same W is added to all coordinates,

$$\max_i X_i = \sqrt{a_N}\,\max_i \xi_i^{(N)} + \frac{W}{\sqrt{N}}.$$

Hence the event $\{\max_i X_i \le c_N\} \cap \{W \ge -L_N\}$ is contained in $\{\max_i \xi_i^{(N)} \le u_N\}$. Theorem 2 gives

$$\begin{aligned}\mathbb{P}\{\max_i \xi_i^{(N)} \le u_N\} &\ge \mathbb{P}\{\max_i X_i \le c_N\} - \mathbb{P}\{W < -L_N\}\\ &\ge \Phi(c_N)^N - o(1).\end{aligned}$$

It remains to identify the limit. From $N\overline{\Phi}(u_N) \to 1$ and Mills' ratio, $u_N \sim \sqrt{2\log N}$. Moreover,

$$0 \le u_N - c_N = (1-\sqrt{a_N})u_N + N^{-1/4}, \qquad u_N(u_N - c_N) = o(1).$$

The standard Mills-ratio comparison therefore gives $\overline{\Phi}(c_N)/\overline{\Phi}(u_N) \to 1$. Thus $N\overline{\Phi}(c_N) \to 1$, and $\Phi(c_N)^N \to e^{-1}$. This proves the lemma. □

Proof of Theorem 1. Fix arbitrary centers $x_1, \ldots, x_N \in \mathbb{S}^{d-1}$ and let A_i be the positive cone over $C_d(x_i)$. For $Y \sim N(0, I_d)$, radial invariance gives

$$\sigma_{d-1}\left(\bigcup_i C_d(x_i)\right) = \gamma_d\left(\bigcup_i A_i\right), \qquad \gamma_d(A_i) = \frac{1}{N}.$$

Let $t = t_{d,N}$ satisfy $\overline{\Phi}(t) = 1/N$. In the proof of its Theorem 1, the source paper constructs a positive trim s_d and $u = t + s_d$ satisfying

$$\sqrt{\log N}\,s_d \to 0, \qquad \overline{\Phi}(u) = \frac{1+o(1)}{N},$$

and proves, uniformly in the cap axis x, that for $H(x) = \{y : \langle y, x\rangle \ge u\}$,

$$\gamma_d\big(A(x) \cap H(x)\big) = \frac{1+o(1)}{N}.$$

Since $\gamma_d(A(x)) = 1/N$ and $\gamma_d(H(x)) = \overline{\Phi}(u) = (1+o(1))/N$, inclusion–exclusion yields the stronger reformulation

$$\gamma_d\big(A(x) \triangle H(x)\big) = o(1/N),$$

again uniformly in x. Consequently,

$$\left|\gamma_d\left(\bigcup_i A_i\right) - \gamma_d\left(\bigcup_i H(x_i)\right)\right| \le \sum_{i=1}^N \gamma_d(A_i \triangle H(x_i)) = o(1). \tag{1}$$

Now put $G = (\langle x_i, x_j \rangle)_{i,j}$ and $\xi_i = \langle Y, x_i \rangle$. Then G is a possibly singular correlation matrix and

$$\gamma_d\left(\bigcup_i H(x_i)\right) = \mathbb{P}\{\max_i \xi_i \geq u\}.$$

Because $N\overline{\Phi}(u) \to 1$, Lemma 1 implies

$$\mathbb{P}\{\max_i \xi_i < u\} \geq e^{-1} - o(1).$$

(The strict and non-strict inequalities have the same probability because each Gaussian coordinate has a continuous marginal.) Together with (1), this proves, uniformly over all centers,

$$U_N(d) \leq 1 - e^{-1} + o(1).$$

For the reverse inequality, choose $x_1, \dots, x_N$ independently and uniformly on the sphere. For every fixed point $z \in \mathbb{S}^{d-1}$, the events $z \in C_d(x_i)$ are independent and each has probability $1/N$. Fubini's theorem gives

$$\mathbb{E}\,\sigma_{d-1}\left(\bigcup_i C_d(x_i)\right) = 1 - \left(1 - \frac{1}{N}\right)^N.$$

Some deterministic configuration attains at least this expectation, so $U_N(d) \geq 1 - (1 - 1/N)^N \to 1 - e^{-1}$. The two bounds prove the theorem. $\square$

5 Verification and dependency audit

1. **Imported theorem.** Theorem 2.1 of arXiv:2607.14087 states the lower-orthant inequality for every correlation matrix R with $R - J/m \succeq 0$, including singular matrices.

2. **Normalization.** $R_N = ((N-1)/N)G_N + J/N$ has diagonal one and $R_N - J/N = ((N-1)/N)G_N \succeq 0$.

3. **Event inclusion.** The common shift is exact, so no independence between the two events is used; only $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) - \mathbb{P}(B^c)$ is invoked.

4. **Tail stability.** The perturbation obeys $u_N(u_N - c_N) = O(u_N^2/N + u_N N^{-1/4}) = o(1)$, precisely the scale needed for the normal-tail ratio to tend to one.

5. **Geometric uniformity.** The source paper proves its cone estimate for one axis and invokes rotational invariance. All caps have the same area, so the $o(1/N)$ error is uniform and survives summation over N caps.

6. **No rank assumption.** The Gram matrix may have rank smaller than N; both the Gaussian construction and Mulgund's theorem permit this.

6 Literature search, novelty, and limitations

Targeted searches for the two arXiv identifiers together, for the source paper's antipodality phrase together with Mulgund's title, and for partial sphere coverings together with stochastic domination of Gaussian maxima did not reveal an explicit statement of Theorem 1. The result should therefore be read as an agent-identified corollary of a later primary theorem, not as a claim that the Gaussian comparison itself is new.

The proof does *not* extend the dimensional regime. Its spherical input is the source estimate $\log N = o(\sqrt{d})$. It also gives no monotonicity of the antipodal quantity $V_N(d)$ and no value of the Laguerre–Dirichlet–Voronoi tiling constant. Those questions remain open on the evidence reviewed here.

7 Human-review checklist

Before external use, a reviewer should check: (i) Theorem 2.1 in the attached supporting paper against the imported statement; (ii) the three source-paper estimates used to derive the symmetric-difference bound; (iii) the direction of the common-noise event inclusion; and (iv) the Mills-ratio perturbation. The proof is short enough that these four checks cover every non-elementary dependency.

8 References

- S. Hoehner and G. Kur, *On the Optimality of Random Partial Sphere Coverings in High Dimensions*, arXiv:2501.10607, latest version consulted August 10, 2026.
- A. Mulgund, *Stochastic Domination of Gaussian Maxima: A Resolution of the Weak Simplex Conjecture*, arXiv:2607.14087v2, July 20, 2026.